Smart Clinic AI Assistant

Project Proposal

Initiative: DEPI — Digital Egypt Pioneers Initiative **Track:** AI & Data Science **Team Leader:** Ahmed Khaled Abdelfattah

---

1. Project Overview

Smart Clinic AI Assistant is an n8n-based automation system that manages a clinic's communication with patients and streamlines internal operations. The system integrates a Telegram chatbot, an AI agent that answers free-text questions, Google Calendar for scheduling, and Google Sheets as the data store, for a single-doctor clinic in the MVP. It reduces manual work for clinic staff, improves the patient experience with 24/7 availability, and gives the doctor(s) an accurate, automated view of daily clinic performance.

2. Project Objectives

1. **Automated Patient Inquiries** — Answer recurring patient questions (location, fees, working hours) via Telegram, around the clock, using button menus for common questions and an AI agent for free-text questions.
2. **Appointment Booking** — Let patients self-book appointments with real-time availability checks against the calendar.
3. **Automated Reminders** — Send an interactive reminder 24 hours before each appointment (Confirm / Reschedule / Cancel) to reduce no-shows.
4. **Financial Reporting** — Generate an automated daily report for the doctor(s) with total revenue and case count, calculated programmatically (not by the LLM) to guarantee accuracy.
5. **Patient Satisfaction Analysis** — Collect post-visit feedback via Telegram, classify it (Positive / Neutral / Negative), and surface it to support service improvement.

3. Project Scope

In scope for the MVP:

- A Telegram-based conversational interface for patients.
- Button-driven menu (Telegram inline keyboard) to route patients between general inquiries and booking requests; booking is fully button-driven and decided by code, with no LLM involved.
- An AI agent that answers free-text questions using clinic information only (it does not book appointments and gives no medical advice).
- Conversation state kept per patient, so name and age can be sent as a normal message, not only as a reply.
- Appointment booking and availability checks against Google Calendar (single doctor, 20-minute slots).
- 24-hour interactive appointment reminders with status tracking.
- A daily, code-calculated financial summary emailed to the doctor(s).
- Post-visit feedback collection and sentiment classification.
- Data storage via Google Sheets (patients, appointments, daily ledger, feedback).

Out of scope for the MVP (candidate future work):

- Migration to a relational database (e.g., PostgreSQL) if data volume or concurrency needs grow.
- WhatsApp integration (Telegram is used for the MVP due to lower setup overhead).
- Handling of non-text patient messages (voice notes, images) beyond a basic fallback reply.
- Multi-branch / multi-clinic support.

4. Target Users

- **Patients** — interact with the system via Telegram to ask questions, book appointments, respond to reminders, and give feedback.
- **Doctor(s)** — receive the daily financial report and benefit from fewer no-shows and better visibility into clinic performance.
- **Clinic secretary/staff** — continue to record payments and attendance in the shared ledger that the system reads from.

5. Proposed Solution

The system is built as a set of n8n workflows orchestrating a Telegram bot (button-driven menus), Google Calendar, Google Sheets, Gmail, an AI agent for free-text questions, and an LLM that classifies post-visit feedback sentiment:

Module	Trigger	Summary
1. Message Handling & Booking	Telegram message	Routes the patient via Telegram inline-button menus; an AI agent answers free-text questions (clinic information only, no booking); answers inquiries with fixed clinic info; remembers the chosen slot per patient so name and age can be sent as a normal message; books appointments against calendar availability (decided by code, never by the LLM)
2. Reminders & Follow-up	Daily/periodic schedule	Sends a 24h-ahead interactive reminder; updates booking status based on patient response (Confirmed / Cancelled / Needs Rescheduling)
3. Daily Financial Report	End-of-day schedule	Reads the day's ledger, computes totals via code (not LLM), and emails a summary to the doctor(s)
4. Patient Feedback & Sentiment	Scheduled after appointment	Requests feedback via Telegram, classifies sentiment, and logs it for review

Data store: Google Sheets is used for the MVP (Bookings, Daily Ledger, Feedback) for fast setup and team accessibility, with a logical data model designed so it can be migrated to a relational database later without a redesign.

6. Assumptions & Open Items

- Clinic scope confirmed: single doctor, 20-minute appointment slots. Cancellation/reschedule policy defaults to 3 hours before the appointment, pending confirmation with the client/instructor.
- Deployment target (working prototype vs. production-ready system) to be confirmed.
- Telegram is assumed as the patient-facing channel for the MVP; WhatsApp is a possible future direction.
- The AI agent is limited to clinic information, has no booking tools, and gives no medical advice; the chat model and system prompt will be documented in the technical documentation.

Project Plan

Timeline (Milestones)

Aligned to the official DEPI submission dates for this track.

Milestone	Deliverables Due	Date
M1 - Planning & Management	Project Proposal, Project Plan, Task Assignment, Risk Assessment, KPIs	16 Oct 2026
M2 - Requirements & Literature Review	Stakeholder Analysis, User Stories, Functional/Non-functional Requirements, Literature Review	16 Oct 2026
M3 - System Analysis & Design	Use Case/Sequence/DFD diagrams, ER Diagram & Schema, Wireframes, Architecture	6 Nov 2026
M4 - Implementation	Working n8n workflows, source code/export on GitHub, README	30 Nov 2026
M5 - Final Presentation & Reports	User Manual, Technical Documentation, Presentation, Infographic	4 Dec 2026

Gantt Overview (by week)

Phase	Weeks	Focus
Requirements & Design	Weeks 1-3 (to 6 Nov)	Confirm clinic requirements, finalize scope, produce all System Analysis & Design artifacts
Core Build	Weeks 4-6	Module 1 (Telegram Assistant) and Module 2 (Booking)
Extended Build	Weeks 6-7	Module 3 (Reminders) and Module 4 (Financial Report)
Final Build & QA	Week 7 (to 30 Nov)	Module 5 (Feedback), end-to-end testing, bug fixes
Wrap-up	Week 8 (to 4 Dec)	Documentation, presentation, infographic

Resource Allocation

- Team Leader (self): planning, architecture decisions, documentation review, integration oversight.
- Member 2: Telegram bot + AI agent and conversation-state logic.
- Member 3: n8n workflow automation (Calendar, Sheets, Gmail integrations).
- Members 4 & 5: testing, bug tracking, documentation support, presentation.

Pending confirmation of each member's actual skills and availability - see Task Assignment below.

Task Assignment & Roles

Member	Primary Responsibility	Key Deliverables
Ahmed Khaled Abdelfattah (Team Lead)	Planning, architecture, documentation, integration review	All Planning & Management docs, final architecture sign-off
Ahmed Hossam Abdullah	n8n automation & integrations (Modules 2-3)	Router/classifier, inquiry & sentiment prompts

Member	Primary Responsibility	Key Deliverables
Mohamed Alaa Ibrahim	Telegram bot & LLM logic (Modules 1, 4-5)	Calendar/Sheets booking and reminder workflows
Ziad Ahmed Fathy	Testing & QA	Test cases, bug reports
Ahmed Mossa Anwar	Documentation & presentation support	User manual, slides, infographic

This split is a starting proposal and will be adjusted once each member confirms their technical background and available hours.

Risk Assessment & Mitigation Plan

Risk	Likelihood	Impact	Mitigation
Low team member engagement	Medium	High	Confirm availability early; redistribute tasks among active members if needed
Delays building n8n workflows	Medium	Medium	Start with a minimal working version per module before adding features
Integration issues between services	Medium	Medium	Test each integration in isolation before connecting modules
Double-booked appointments	Low	High	Availability check always goes through Calendar/Code Node, never the LLM alone
Financial report errors	Low	High	All calculations done in a Code Node, not by the LLM
Inaccurate AI responses	Medium	Medium	The AI agent answers from clinic information only, has no booking tools, gives no medical advice, and falls back to the menu if it fails; business logic stays in code
Patient data protection	Medium	High	Limit data collected to what's needed; restrict booking actions to the requesting patient's own chat_id

Risk	Likelihood	Impact	Mitigation
Missing the final deadline	Low	High	Follow the milestone schedule above; flag slippage early to the team

KPIs (Key Performance Indicators)

KPI	Target	How It's Measured
Response Time	Reply within a few seconds of a patient message	Timestamp difference, Telegram message to bot reply
Inquiry Resolution Rate	High share of inquiries answered without human handoff	Ratio of inquiries answered by Module 1 vs. escalated
Booking Success Rate	High share of booking attempts completed successfully	Confirmed bookings / total booking attempts
Reminder Delivery Rate	Reminders sent for all confirmed appointments	Reminders sent / appointments due
Financial Report Accuracy	Report totals match the ledger exactly	Automated total vs. manual spot-check
Feedback Collection Rate	Reasonable share of patients respond to the feedback request	Feedback responses / feedback requests sent
Milestone Completion Rate	All milestones delivered on or near their due date	Milestones completed on time / total milestones

Exact target percentages to be set once the MVP is running and we have baseline numbers from testing.